

VERKHOFYANSK, Russian Federation

WMO: 242660

Lat: 67.565N

Lon: 133.412E

Elev: 447

StdP: 14.46

Time Zone: 10.00 (E10)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-68.1	-65.3	-70.7	0.1	-66.3	-67.3	0.1	-64.1	8.1	-39.6	7.1	-42.7	1.4	240	0.796

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	84.0	64.3	79.7	62.3	75.3	60.2	65.9	81.3	63.6	77.1	61.2	73.2	5.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	78.0	72.3	57.4	71.5	71.2	54.9	65.1	68.3	31.0	81.2	29.2	77.1	27.5	73.4	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.2	12.1	10.0	DB	-69.3	89.8	3.9	4.4	-72.1	92.9	-74.4	95.5	-76.6	97.9	-79.5	101.1	
			WB	-69.5	68.8	3.9	3.7	-72.3	71.5	-74.5	73.6	-76.7	75.7	-79.5	78.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-49.6	-43.2	-18.9	10.9	38.1	55.5	60.1	53.7	37.1	6.1	-29.7	-45.8
	DBStd	42.28	11.35	11.55	15.52	13.75	10.09	8.71	8.76	7.53	7.84	16.26	14.79	12.06
	HDD50	16615	3088	2610	2137	1172	383	45	16	46	393	1362	2392	2971
	HDD65	21445	3553	3030	2602	1622	834	301	201	359	838	1827	2842	3436
	CDD50	719	0	0	0	0	14	210	330	160	5	0	0	0
	CDD65	75	0	0	0	0	0	17	50	8	0	0	0	0
	CDH74	1173	0	0	0	0	4	312	707	150	0	0	0	0
	CDH80	337	0	0	0	0	0	76	230	31	0	0	0	0
	WSAvg	3.2	1.5	1.3	1.9	3.5	5.4	5.7	5.0	3.9	3.7	2.9	1.9	1.7
	PrecAvg	7.1	0.2	0.2	0.2	0.2	0.5	1.1	1.3	1.1	0.7	0.5	0.4	0.3
Precipitation	PrecMax	11.3	0.7	0.9	0.6	0.8	1.8	3.6	4.5	2.7	2.1	1.5	1.6	1.2
	PrecMin	4.2	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.2	0.1	0.1	0.1	0.0
	PrecStd	1.8	0.1	0.2	0.1	0.2	0.4	0.7	0.9	0.6	0.5	0.3	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.0	4.6	33.1	48.1	72.0	86.9	91.3	84.3	66.9	41.5	15.8	-2.4
		MCWB	-7.5	3.1	27.8	38.8	54.9	65.8	68.6	64.5	53.1	34.7	13.6	-3.1
	2%	DB	-19.3	-11.0	24.7	42.1	65.2	81.7	86.1	78.3	59.6	35.2	4.1	-12.2
		MCWB	-19.5	-11.4	20.6	34.6	50.7	62.4	66.2	61.6	49.0	30.4	3.3	-12.6
	5%	DB	-27.0	-20.0	16.8	37.8	60.2	77.4	81.9	73.2	54.8	31.8	-1.6	-21.2
		MCWB	-27.2	-20.2	14.1	31.6	47.8	59.4	64.0	59.3	46.2	28.5	-2.1	-21.5
	10%	DB	-33.6	-26.9	7.9	33.5	55.6	72.6	77.7	68.7	50.2	28.3	-7.4	-27.8
		MCWB	-33.8	-27.1	6.2	28.4	45.2	57.4	62.0	56.8	43.4	25.6	-7.9	-28.0
	0.4%	WB	-7.4	3.0	27.8	39.0	55.2	66.3	70.0	66.3	53.9	35.2	13.6	-2.8
		MCDB	-7.1	4.7	32.8	47.8	71.3	84.3	87.2	81.5	64.8	40.3	15.7	-2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-19.5	-11.4	20.8	35.2	51.4	63.4	67.3	63.0	50.0	31.3	3.4	-12.6
		MCDB	-19.3	-11.0	24.6	41.5	63.7	80.1	83.4	75.2	58.1	34.7	4.1	-12.2
	5%	WB	-27.3	-20.2	14.1	31.8	48.4	60.6	65.2	60.4	46.8	28.5	-2.2	-21.3
		MCDB	-27.1	-20.0	16.8	37.5	59.4	74.9	79.2	71.1	53.9	31.7	-1.5	-21.0
	10%	WB	-33.7	-27.1	6.2	28.5	45.5	58.1	63.0	57.9	43.9	25.5	-7.9	-28.0
		MCDB	-33.6	-26.9	7.9	33.6	55.2	71.1	75.5	67.3	49.5	28.4	-7.3	-27.8
Mean Daily Temperature Range	5% DB	MDBR	9.6	15.5	30.5	30.2	19.2	20.4	20.5	20.8	17.2	16.2	11.8	10.0
		MCDBR	19.0	22.3	32.7	25.2	27.4	29.1	29.7	29.7	25.5	15.5	17.5	21.7
		MCWBR	18.7	21.9	29.2	19.0	15.8	13.8	13.2	14.7	15.5	11.9	16.8	21.3
	5% WB	MCDBR	18.9	22.3	32.4	24.0	25.3	27.0	26.3	26.1	23.3	14.4	17.7	21.6
		MCWBR	18.7	21.9	28.9	18.3	14.9	13.6	12.6	13.2	14.6	11.3	17.0	21.3
Clear-Sky Solar Irradiance	taub		0.173	0.226	0.287	0.337	0.379	0.377	0.469	0.420	0.305	0.243	0.178	N/A
	taud		1.887	2.060	2.057	2.015	2.182	2.369	2.065	2.180	2.479	2.213	1.909	N/A
	Ebn at Noon		48	190	234	252	256	262	226	225	239	182	40	0
	Edh at Noon		3	16	29	40	38	33	43	35	20	13	3	0
All-Sky Solar Radiation	RadAvg		16	223	764	1485	1734	1800	1630	1218	667	268	40	0
	RadStd		2	11	41	64	117	161	99	102	46	31	4	0

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.81	+1.04	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page